

SINGLE WIRE DEBUG
USB INTERFACE TO TOP IO BOARD
J14+12V TO CTRL BOARD FROM IO BOARD
SSW-110-01-F-D

NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,
DISABLE TIM1 SYNC PIN (TIM1 ETR)

Connector from gate drivers

Includes multiple redundant SSO (six-switch-off) pathways:

- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2

(Gate driver power enable implements 1002, with monitoring feedback).

feedback pins are 0-3.3v binary (3.3v expected when +12v present)

Connector to IO board
Digital Outputs provided here to IOBoard.
J7

SSW-108-01-L-D

HALL_U_ENCODER_SIN_IN 1 16 DOUT4_A
+5V_A 2 15 DOUT3_A
HALL_V_ENCODER_COS_IN 3 14 DOUT2_A
GND_A 4 13 DOUT1_A
HALL_W_IN 5 12 GND_A
GND_A 6 11 THROTTLE_B_IN
MOTOR_TEMP_IN 7 10 GND_A
GND_A 8 9 THROTTLE_A_IN

GND_A 1 20 +12V_IN_A
MP_SWO 2 19 GND_A
MP_BOOTSEL 3 18 CP_BOOTSEL
MP_SWCLK 4 17 CP_SWCLK
MP_SWIDIO 5 16 CP_SWIDIO
MP_RESET 6 15 CP_RESET
GND_A 7 14 +3.3V_A
USB_DP 8 13 USB_DM
MD_FIBER_TX 9 12 MD_FIBER_SYNC
MD_FIBER_RX 10 11

This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)

Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.

TemperatureSensors

HS_TMP_3 HEATSINK_TMP2_V
HS_TMP_2 HEATSINK_TMP3_V
HS_TMP_1 HEATSINK_TMP_V
MTR_TMP_1 MOTOR_TEMP_IN
+3.3V_V
GND_V

File: TemperatureSensors.kicad_sch

CONNECTOR FROM THIS GATE DRIVER
TO MAIN CONTROL BOARD, BELOW

Voltage Divider Network3

H1 971350581

H2 971350581

H3 971350581

H4 971350581

H5 971350581

H6 971350581

H7 971350581

H8 971350581

H9 971350581

H10 971350581

H11 971350581

H12 971350581

H13 971350581

H14 971350581

H15 971350581

H16 971350581

H17 971350581

H18 971350581

H19 971350581

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H223 971350581

H224 971350581

H225 971350581

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H233 971350581

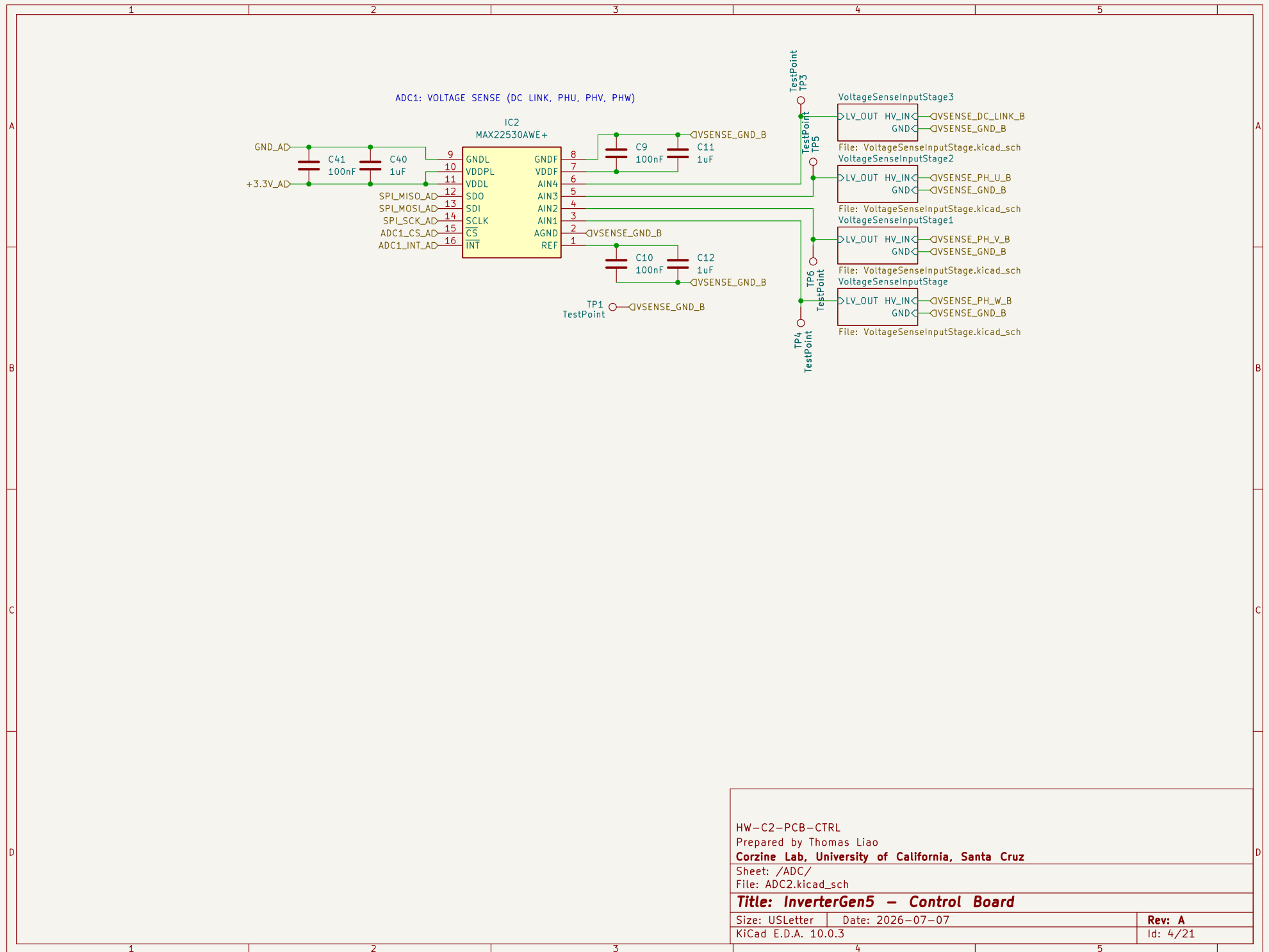
H234 971350581

H235 971350581

H236 971350581

H237 971350581

H238 9



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/

File: ADC2.kicad_sch

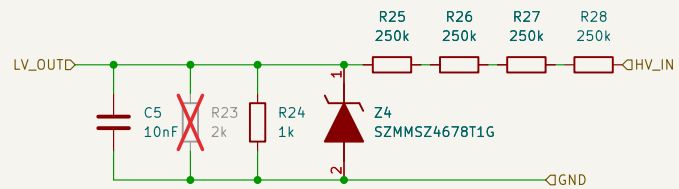
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

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Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad_sch

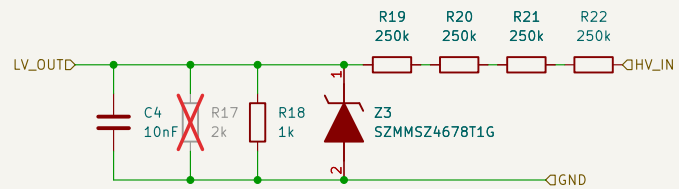
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad_sch

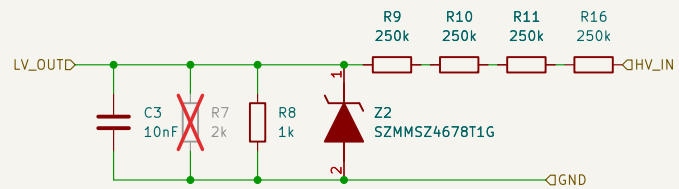
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

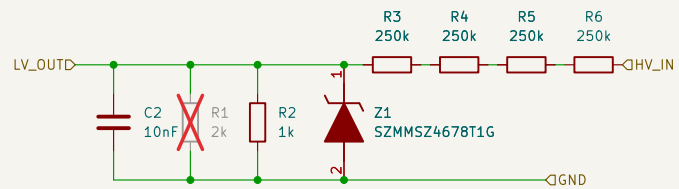
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

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Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

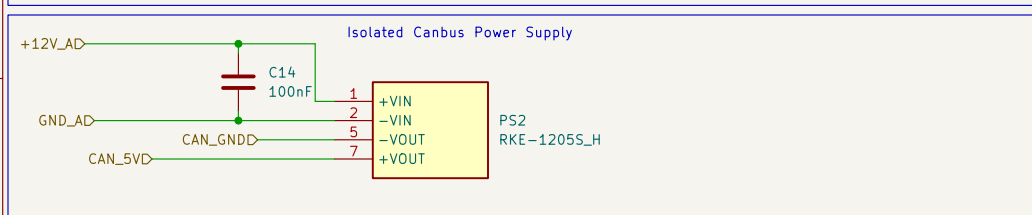
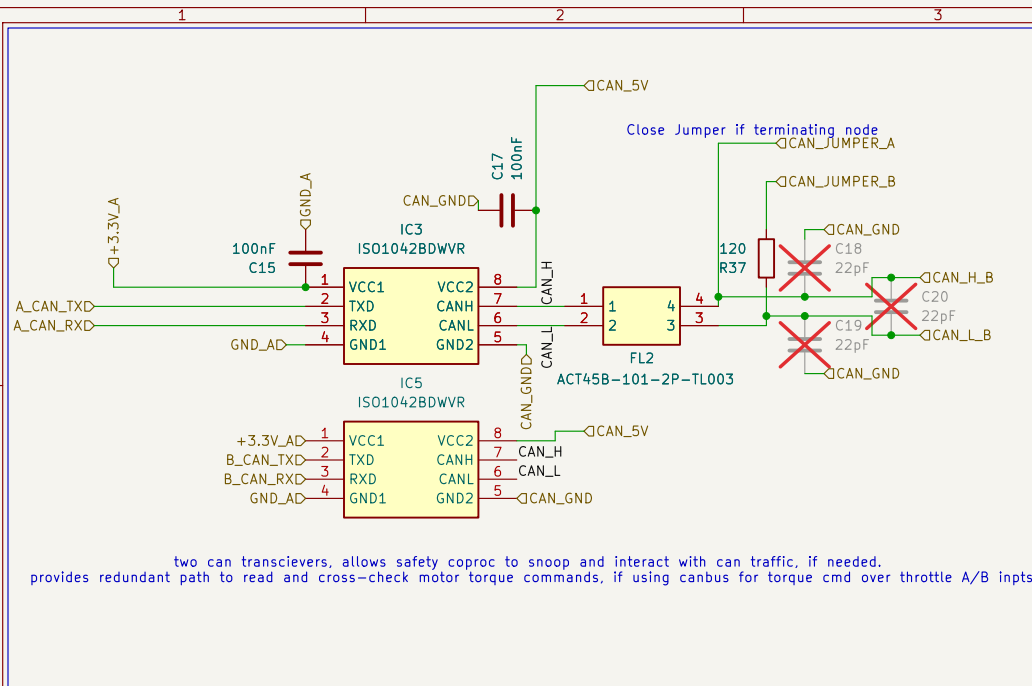
Size: USLetter

Date: 2026-07-07

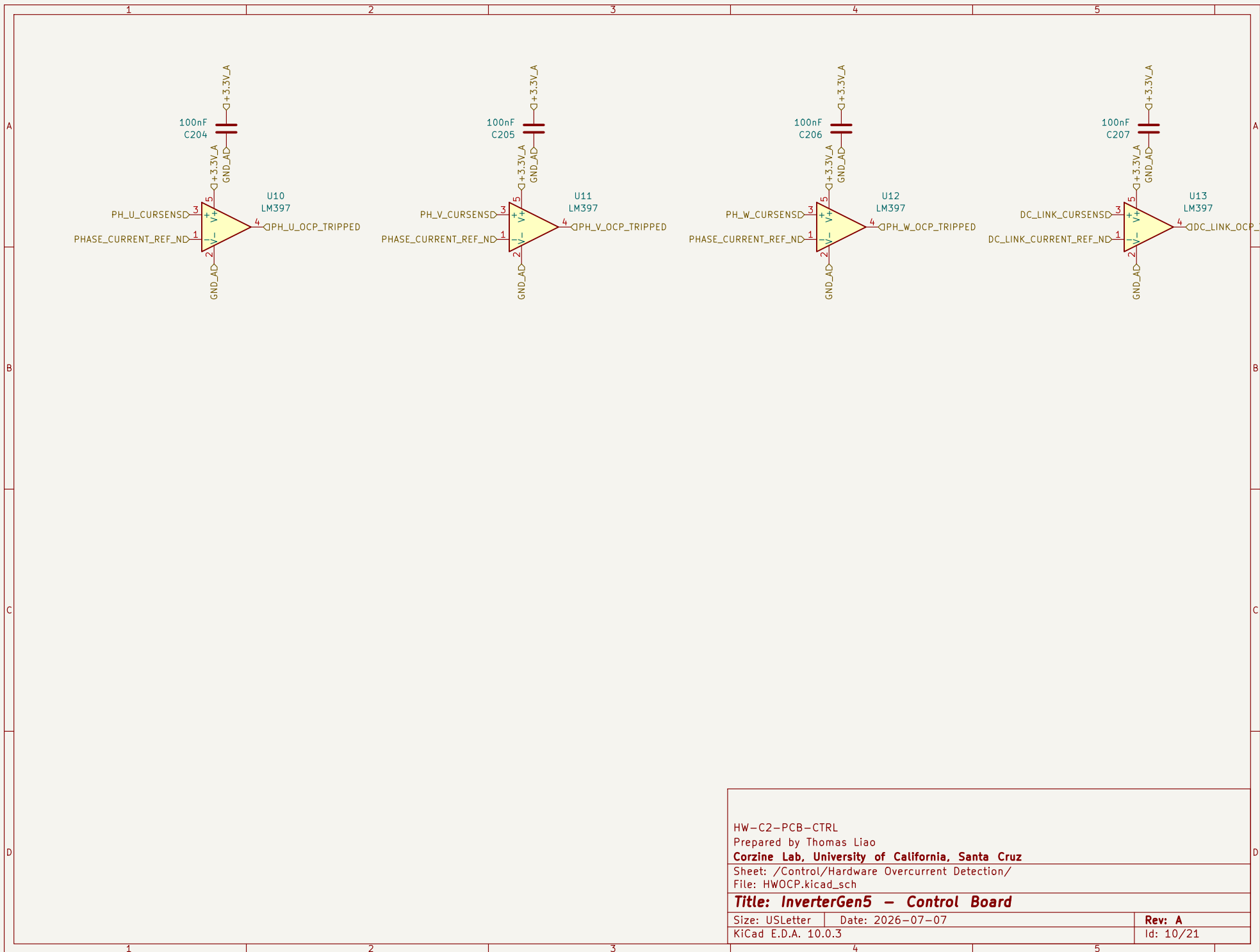
Rev: A

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Id: 6/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever1/
File: CANTransciever.kicad_sch
Title: InverterGen5 – Control Board
Size: USLetter | Date: 2026-07-07
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HW-C2-PCB-CTRL

Prepared by Thomas Liao

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Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

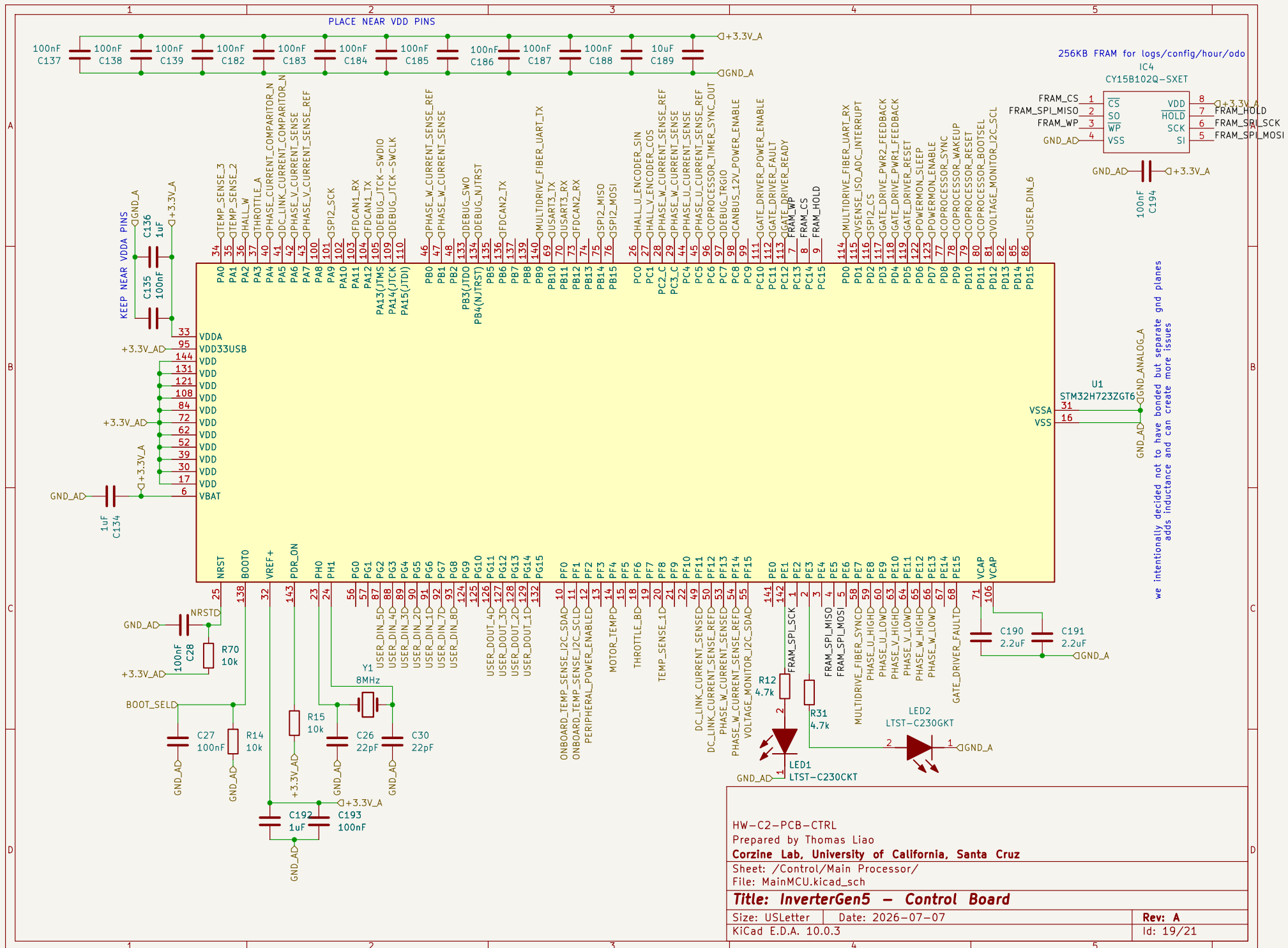
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Size: USLetter Date: 2026-07-07

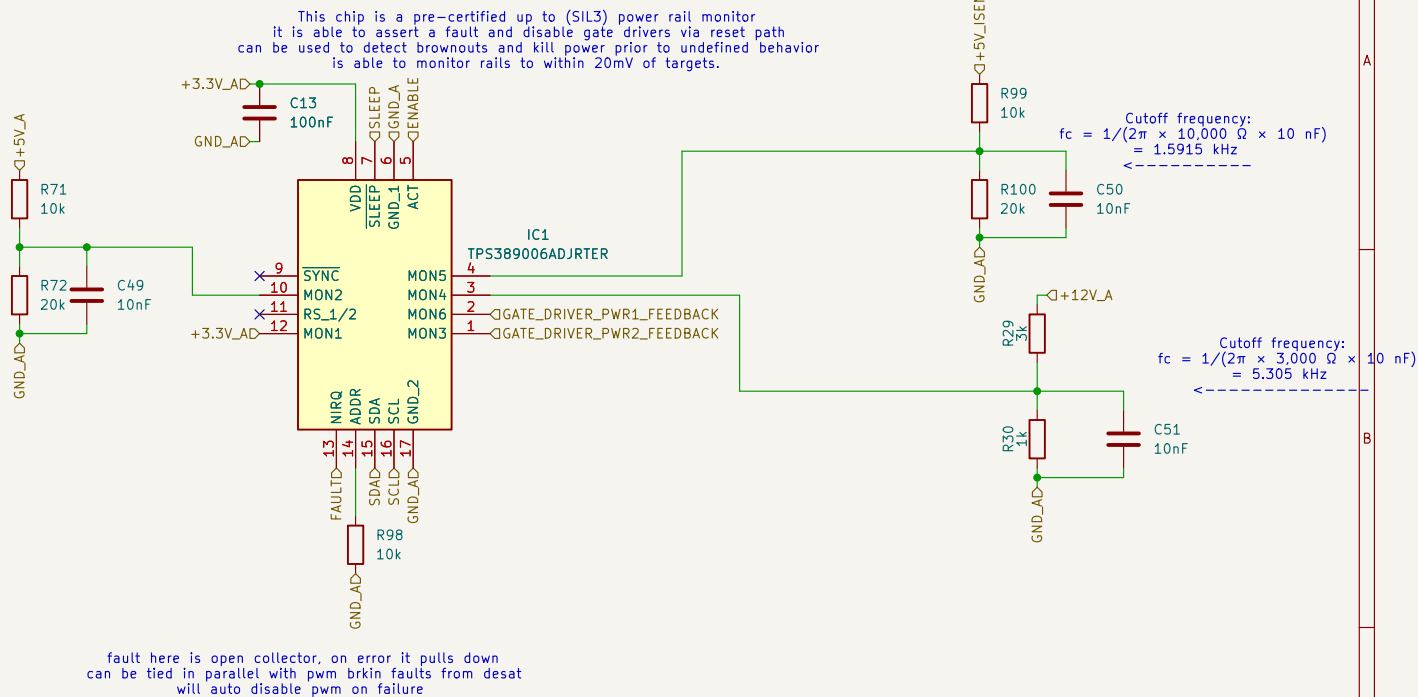
KiCad E.D.A. 10.0.3

Rev: A

Id: 10/21



Cutoff frequency:
 $f_c = 1/(2\pi \times 10,000 \Omega \times 10 \text{ nF})$
 $= 1.5915 \text{ kHz}$



HW-C2-PCB-CTRL

Prepared by Thomas Liao

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Sheet: /Control/Power Supply Monitor/

File: PowerSupplyMonitor.kicad_sch

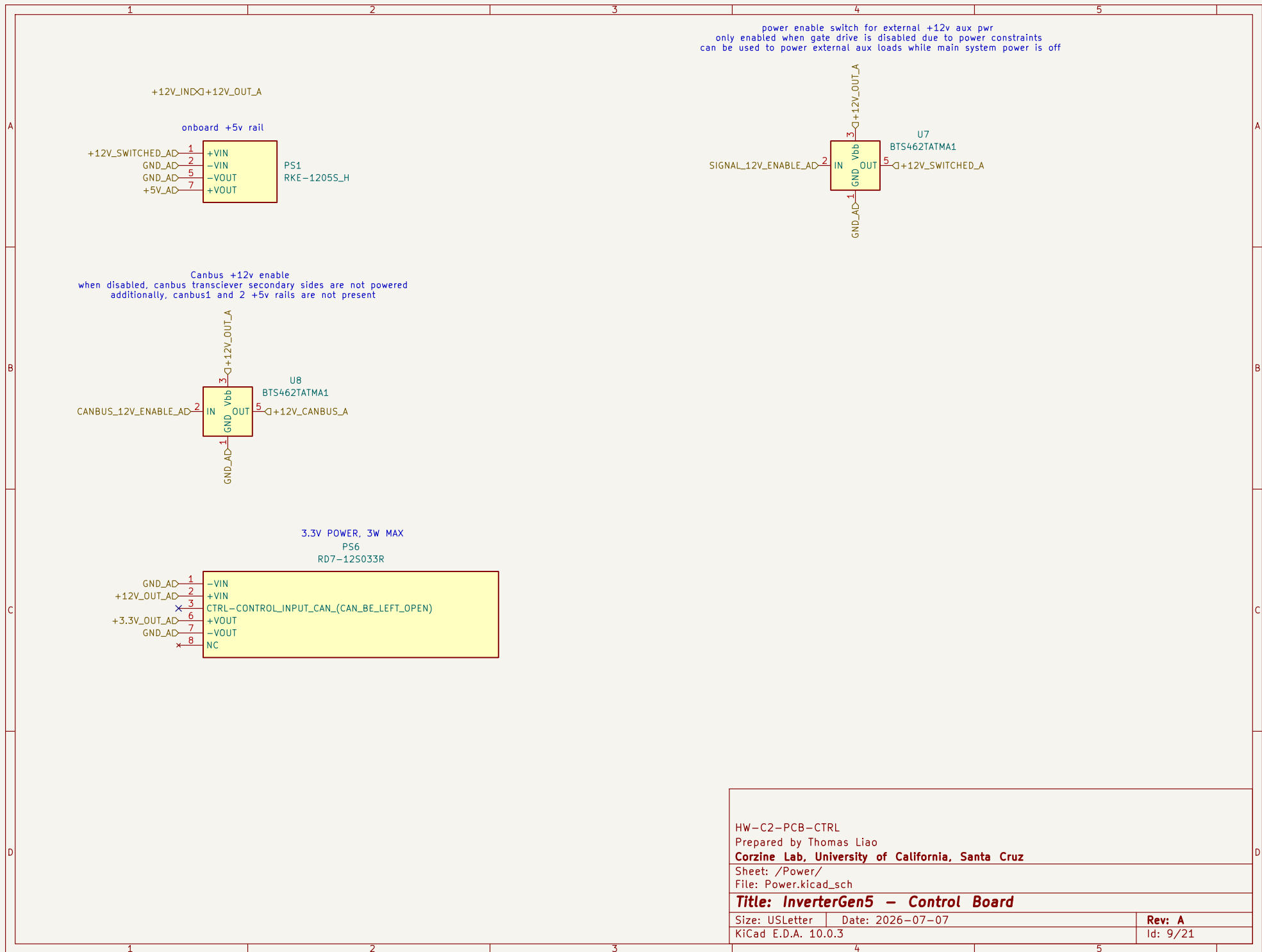
Title: InverterGen5 - Control Board

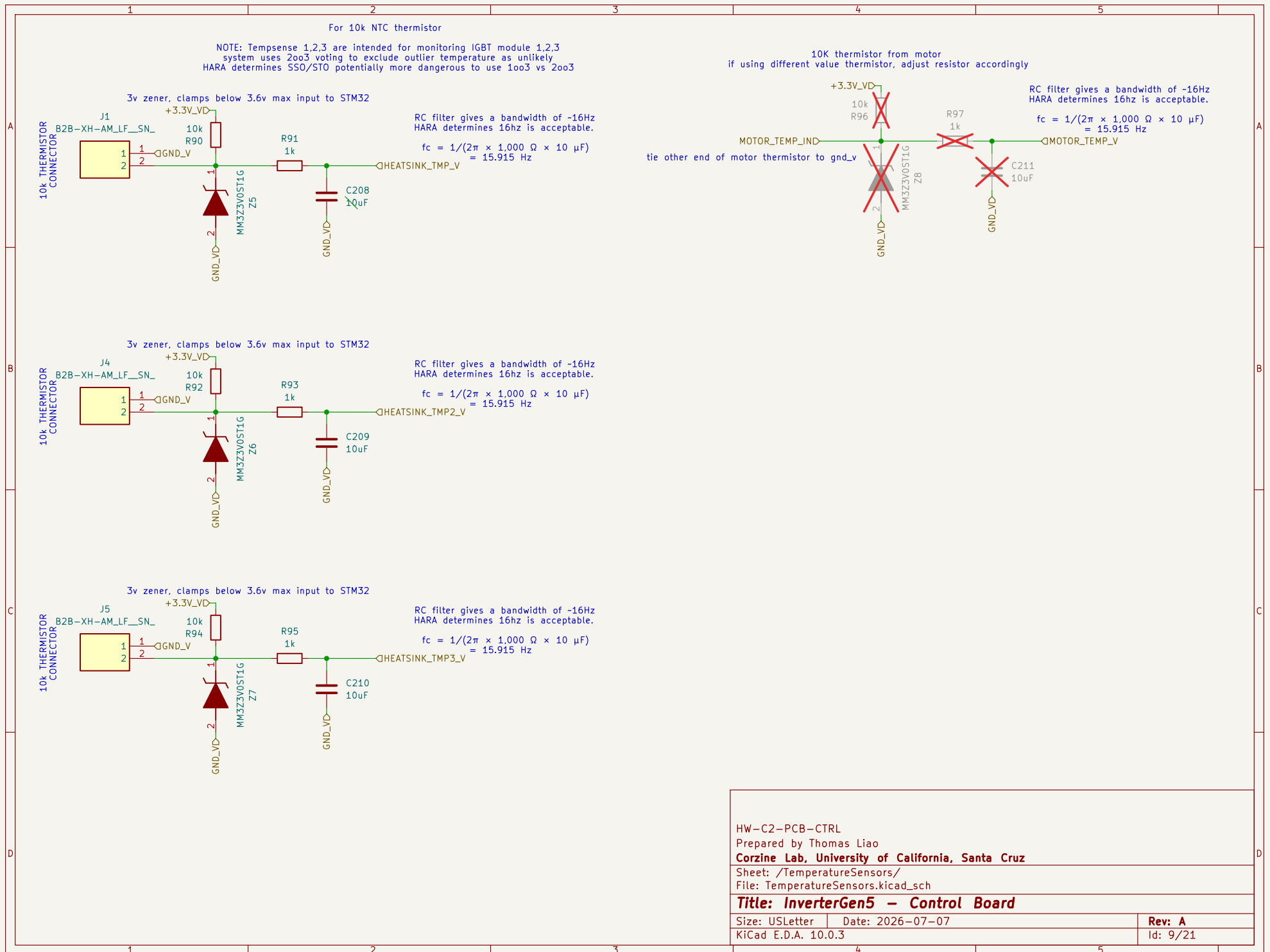
Size: USLetter Date: 2026-07-07

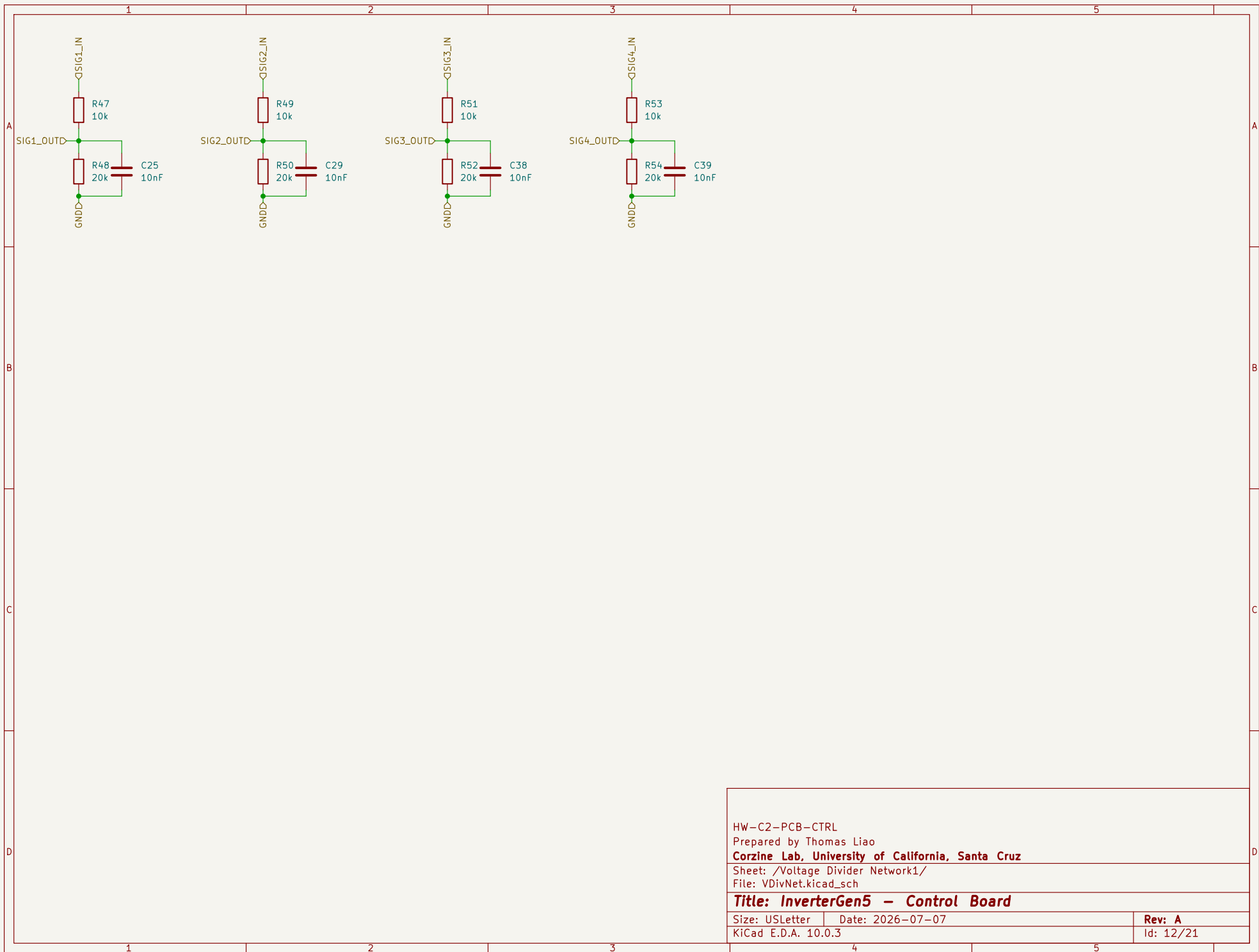
KiCad E.D.A. 10.0.3

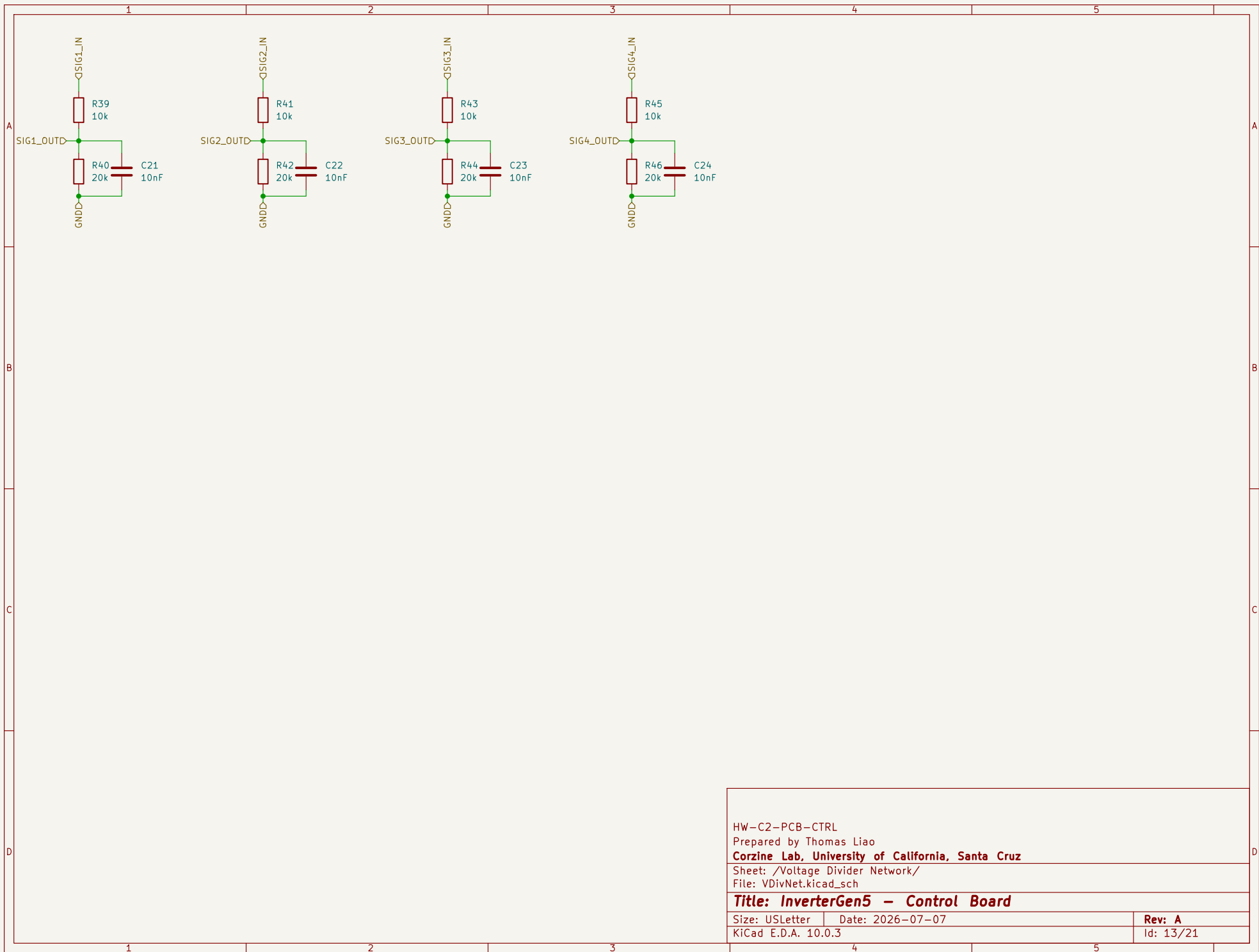
Rev: A

Id: 20/21









HW-C2-PCB-CTRL

Prepared by Thomas Liao

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Sheet: /Voltage Divider Network/

File: VDivNet.kicad_sch

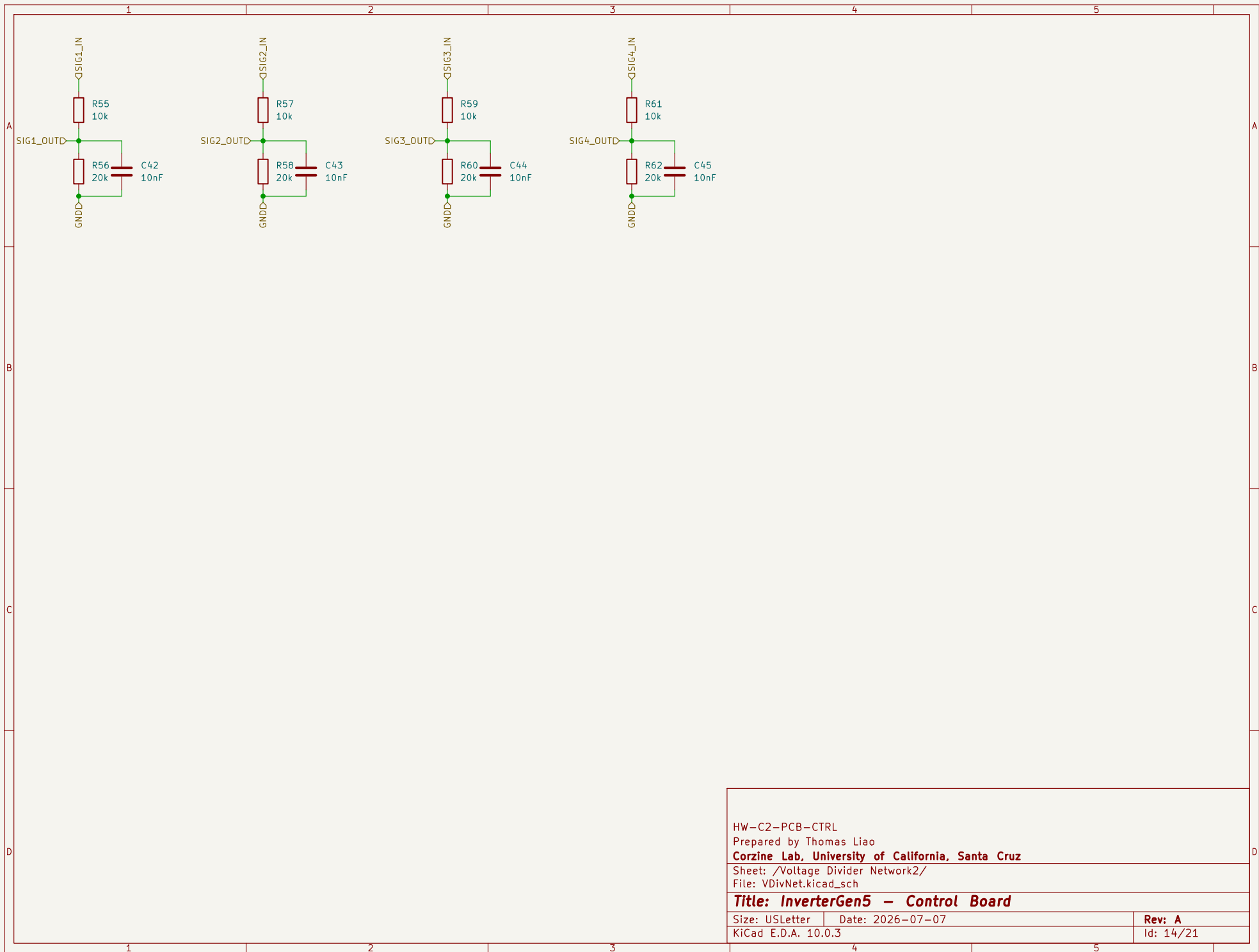
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

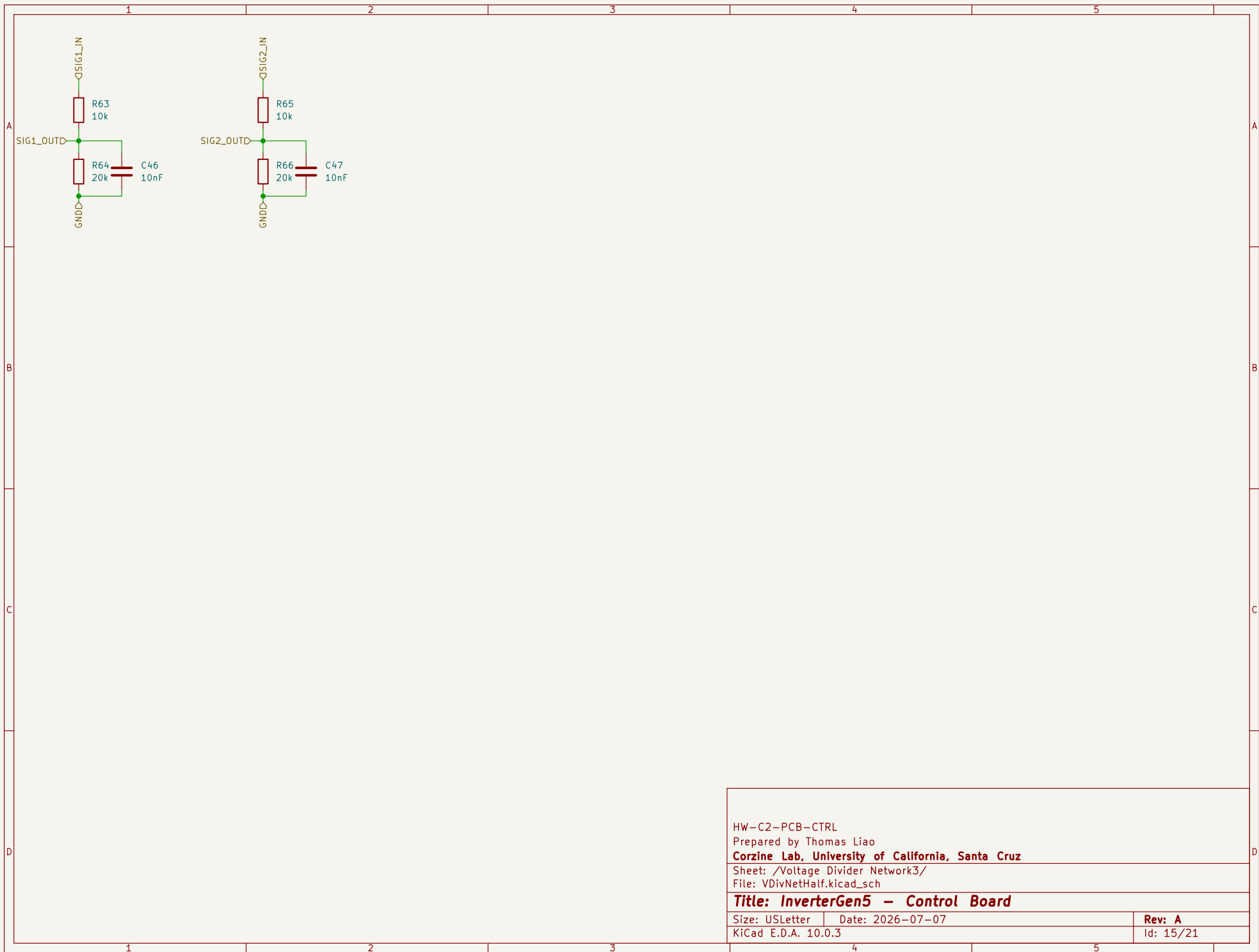
KiCad E.D.A. 10.0.3

Rev: A

Id: 13/21



HW-C2-PCB-CTRL		
Prepared by Thomas Liao		
Corzine Lab, University of California, Santa Cruz		
Sheet: /Voltage Divider Network2/		
File: VDivNet.kicad_sch		
Title: InverterGen5 - Control Board		
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 14/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network3/

File: VDivNetHalf.kicad_sch

Title: InverterGen5 - Control Board

Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 15/21

